

## VOLCHIKHA, Russian Federation

WMO: 360220

Lat: 52.022N

Lon: 80.340E

Elev: 683

StdP: 14.34

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | -36.6      | -29.5 | -41.7                         | 0.5 | -36.3 | -34.6 | 0.8 | -29.3 | 24.6                  | 19.6 | 21.6 | 20.5 | 3.0                   | 40   | 0.647 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 21.3                   | 89.3            | 65.0 | 85.9 | 64.0 | 82.7 | 63.0 | 68.5                | 81.3 | 66.9 | 80.0 | 65.5 | 78.1 | 9.5                  | 60   |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 64.6                            | 93.5 | 72.8 | 62.7 | 87.3 | 70.9 | 61.0 | 82.1 | 70.4 | 33.3          | 81.5 | 31.9 | 79.7 | 30.8 | 78.6 | 84.9              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |       |                    |     | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |       |     |
|-------------------|------|------|----------------------------|-------|--------------------|-----|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|-------|-----|
|                   |      |      | Mean                       |       | Standard Deviation |     | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       |       |     |
| 1%                | 2.5% | 5%   | Min                        | Max   | Min                | Max | Min                                                | Max   | Min        | Max   | Min        | Max   | Min        | Max   |       |     |
| (a)               | (b)  | (c)  | (e)                        | (f)   | (g)                | (h) | (i)                                                | (j)   | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |       |     |
| (4) 22.8          | 19.8 | 17.4 | DB                         | -40.7 | 95.7               | 7.2 | 3.2                                                | -45.9 | 98.0       | -50.1 | 99.8       | -54.1 | 101.6      | -59.3 | 103.9 | (4) |
| (5)               |      |      | WB                         | -40.7 | 71.5               | 6.9 | 3.5                                                | -45.7 | 74.0       | -49.7 | 76.0       | -53.6 | 78.0       | -58.6 | 80.5  | (5) |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |  |  |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--|--|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |  |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 37.0   | 0.4   | 5.0   | 20.2  | 41.8  | 54.8  | 65.5  | 68.1  | 64.0  | 52.5  | 39.5  | 21.7  | 8.0   |  |  |
|                                                                        | DBStd       | 26.40  | 16.40 | 14.75 | 12.92 | 10.67 | 8.96  | 6.98  | 5.92  | 6.72  | 8.22  | 9.17  | 14.70 | 14.84 |  |  |
|                                                                        | HDD50       | 6621   | 1538  | 1260  | 924   | 284   | 55    | 1     | 0     | 1     | 66    | 342   | 848   | 1302  |  |  |
|                                                                        | HDD65       | 10554  | 2003  | 1680  | 1389  | 698   | 333   | 79    | 36    | 99    | 381   | 791   | 1298  | 1767  |  |  |
|                                                                        | CDD50       | 1860   | 0     | 0     | 0     | 37    | 204   | 465   | 561   | 436   | 142   | 15    | 0     | 0     |  |  |
|                                                                        | CDD65       | 317    | 0     | 0     | 0     | 1     | 16    | 92    | 132   | 69    | 7     | 0     | 0     | 0     |  |  |
|                                                                        | CDH74       | 4131   | 0     | 0     | 0     | 33    | 355   | 1154  | 1365  | 1016  | 207   | 1     | 0     | 0     |  |  |
|                                                                        | CDH80       | 1329   | 0     | 0     | 0     | 6     | 95    | 369   | 450   | 349   | 60    | 0     | 0     | 0     |  |  |
| (14) Wind                                                              | WSAvg       | 8.0    | 7.5   | 7.7   | 8.5   | 9.2   | 9.1   | 7.9   | 7.1   | 7.2   | 7.3   | 7.9   | 8.4   | 8.2   |  |  |
| (15) Precipitation                                                     | PrecAvg     | 13.8   | 0.7   | 0.6   | 0.8   | 0.8   | 1.0   | 1.6   | 2.4   | 1.3   | 1.0   | 1.2   | 1.3   | 1.0   |  |  |
|                                                                        | PrecMax     | 18.6   | 1.4   | 1.3   | 2.8   | 2.2   | 3.6   | 4.8   | 5.7   | 5.0   | 3.0   | 3.3   | 2.5   | 1.9   |  |  |
|                                                                        | PrecMin     | 7.5    | 0.2   | 0.0   | 0.2   | 0.0   | 0.4   | 0.3   | 0.6   | 0.2   | 0.1   | 0.4   | 0.5   | 0.4   |  |  |
|                                                                        | PrecStd     | 2.9    | 0.3   | 0.3   | 0.6   | 0.6   | 0.7   | 1.0   | 1.3   | 0.9   | 0.6   | 0.7   | 0.5   | 0.4   |  |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 35.1  | 35.9  | 53.7  | 78.6  | 88.7  | 91.8  | 93.3  | 94.2  | 86.9  | 71.7  | 53.9  | 36.0  |  |  |
|                                                                        |             | MCWB   | 32.3  | 32.4  | 42.7  | 54.9  | 61.5  | 64.5  | 67.3  | 65.9  | 60.9  | 52.8  | 44.4  | 33.1  |  |  |
|                                                                        | 2%          | DB     | 31.1  | 32.5  | 45.0  | 71.0  | 82.5  | 87.6  | 88.8  | 88.1  | 80.0  | 64.8  | 47.5  | 32.8  |  |  |
|                                                                        |             | MCWB   | 28.8  | 29.7  | 38.6  | 52.4  | 58.6  | 63.8  | 66.7  | 64.6  | 58.4  | 49.8  | 41.3  | 30.8  |  |  |
|                                                                        | 5%          | DB     | 27.0  | 29.7  | 40.2  | 65.5  | 78.1  | 84.3  | 84.9  | 83.7  | 74.3  | 59.5  | 42.5  | 30.2  |  |  |
|                                                                        |             | MCWB   | 24.8  | 27.0  | 35.4  | 49.6  | 57.0  | 63.2  | 65.5  | 63.7  | 55.9  | 47.7  | 37.7  | 28.2  |  |  |
|                                                                        | 10%         | DB     | 23.2  | 26.3  | 36.7  | 60.1  | 73.4  | 80.9  | 81.5  | 79.8  | 69.0  | 54.0  | 38.3  | 27.5  |  |  |
|                                                                        |             | MCWB   | 21.2  | 24.0  | 33.2  | 47.2  | 55.4  | 62.5  | 64.9  | 62.1  | 54.2  | 45.0  | 35.1  | 25.7  |  |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 32.8  | 33.2  | 44.0  | 57.1  | 63.7  | 68.5  | 71.6  | 69.3  | 62.9  | 54.4  | 46.3  | 33.8  |  |  |
|                                                                        |             | MCDB   | 34.7  | 34.9  | 51.4  | 74.1  | 81.2  | 82.1  | 84.3  | 85.0  | 81.5  | 66.8  | 51.4  | 35.4  |  |  |
|                                                                        | 2%          | WB     | 29.1  | 29.9  | 39.0  | 53.9  | 60.9  | 66.6  | 69.5  | 66.7  | 59.6  | 51.3  | 41.4  | 31.1  |  |  |
|                                                                        |             | MCDB   | 30.9  | 32.0  | 44.1  | 67.4  | 77.4  | 80.6  | 80.9  | 82.7  | 75.3  | 62.3  | 46.6  | 32.6  |  |  |
|                                                                        | 5%          | WB     | 24.9  | 27.1  | 35.9  | 50.9  | 58.8  | 65.2  | 68.0  | 65.1  | 57.5  | 48.5  | 38.1  | 28.4  |  |  |
|                                                                        |             | MCDB   | 26.8  | 29.5  | 39.6  | 62.9  | 74.0  | 79.1  | 79.8  | 79.6  | 72.1  | 58.2  | 42.5  | 29.9  |  |  |
|                                                                        | 10%         | WB     | 21.3  | 24.0  | 33.4  | 48.2  | 56.7  | 63.9  | 66.5  | 63.4  | 55.2  | 45.8  | 35.0  | 25.8  |  |  |
|                                                                        |             | MCDB   | 23.1  | 26.3  | 36.3  | 59.2  | 70.5  | 77.2  | 78.2  | 76.6  | 67.3  | 53.1  | 38.2  | 27.4  |  |  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 18.3  | 20.9  | 19.7  | 22.4  | 24.9  | 22.9  | 21.3  | 24.3  | 24.0  | 19.1  | 14.8  | 17.0  |  |  |
|                                                                        |             | MCDBR  | 16.7  | 17.1  | 19.8  | 31.6  | 33.2  | 29.5  | 27.4  | 31.6  | 34.6  | 30.5  | 17.2  | 15.4  |  |  |
|                                                                        |             | MCWBR  | 15.7  | 14.9  | 14.1  | 16.1  | 14.7  | 11.6  | 10.0  | 12.0  | 15.7  | 17.5  | 12.0  | 14.6  |  |  |
|                                                                        |             | MCDBR  | 17.1  | 16.5  | 17.5  | 28.7  | 27.3  | 24.3  | 22.2  | 26.9  | 29.7  | 27.6  | 15.7  | 15.3  |  |  |
|                                                                        | 5% WB       | MCWBR  | 16.4  | 14.8  | 12.9  | 16.0  | 13.4  | 10.5  | 9.4   | 11.4  | 14.6  | 16.7  | 11.5  | 14.9  |  |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.299 | 0.336 | 0.388 | 0.410 | 0.415 | 0.413 | 0.436 | 0.414 | 0.373 | 0.354 | 0.339 | 0.307 |  |  |
|                                                                        | taud        |        | 2.031 | 1.954 | 1.926 | 2.143 | 2.184 | 2.275 | 2.212 | 2.243 | 2.303 | 2.254 | 2.098 | 1.990 |  |  |
|                                                                        | Ebn at Noon |        | 208   | 234   | 246   | 260   | 266   | 267   | 258   | 257   | 254   | 232   | 190   | 177   |  |  |
|                                                                        | Edh at Noon |        | 26    | 37    | 47    | 43    | 43    | 40    | 42    | 39    | 32    | 28    | 24    | 23    |  |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 402   | 757   | 1217  | 1500  | 1899  | 2027  | 1838  | 1613  | 1185  | 640   | 329   | 276   |  |  |
|                                                                        | RadStd      |        | 31    | 38    | 102   | 110   | 137   | 146   | 123   | 96    | 97    | 85    | 39    | 37    |  |  |

## Historical Trends

|                              | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                              |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                              |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)   | (k)   | (l)   |
| (46) Station Only            | N/A   | N/A     | N/A    | N/A     | -0.83 | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) Regional (10 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page